Factors Affecting Crime Rate In the United States

Final Research Project

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Introduction

The FBI defines violent crime as a combination of forcible rape, robbery, aggravated assault, murder and non-negligent manslaughter. The percentage breakdown in 2008 for violent crime is as follows: aggravated assault (60.4%), robbery (32.0%), forcible rate (6.4%) and murder (1.2%). In the United States, an estimated 1,382,012 violent crimes occurred nationwide in 2008. This was a 1.6 percent increase from 2004, but a 3.2 percent decrease from 1999. In 2008, there were an estimated 454.4 violent crimes per 100,000 inhabitants with the crime rate varying greatly from one city to the next. The violent crime rate is an important statistic for cities in many ways. It not only affects the quality of life for the inhabitants, but it also affects the economy. The economic impact is felt in loss of tourism, attraction of businesses or events to the city and thereby affecting the inflow of employed and educated people. The rate of violent crime also reflects the economy. An increase in violent crime quite possibly indicates an increase in unemployment, lower income, broken families, among other factors that are examined in this report. This report tests factors that possibly affect the crime rate in 40 selected cities across the United States.

Each year, millions of dollars and hours of time are used to curb violent crime across the United States and there is much debate is on what these resources are going into. With more knowledge on what factors actually affect violent crime, more effective policy decisions can be made in addressing these factors specifically and less time and money will be wasted on programs that miss the mark on what actually is behind the violent crime rates. This report will attempt to analyze which of these factors correlate to violent crime the most.

Research problem

The research question is this. Why does the violent crime rate vary from one city to the next and what factors influence this variation? This report attempts to answer this question. Many factors are involved in determining why a city has a certain crime rate. This study will perform multiple regression analysis on thirteen factors (independent variables) to see how well each of these correlates to the crime rate (dependent variable) for each of the forty cities studied. The follow are the factors studied in this report: Household Size, Percent High School Graduate or Higher, Percent Bachelors Degree or Higher, Male (now married except separated), In labor force (population 16 years and older), Median household income (2008 Inflation adjusted dollars), Individuals below poverty level, Owner occupied housing units, Median age, Percent white, Percent black, Percent Hispanic or Latino or any race and population.

Data Sources

The crime data is from the Federal Bureau of Investigation (FBI) website and the demographic data for each city are from the U. S. Census Bureau. The data are presented in the 2008 edition of the FBI's annual publication "Crime in the United States". This is a statistical compilation of offense and arrest data as reported by law enforcement agencies that participate voluntarily in the FBI's Uniform Crime Reporting (UCR) Program.

The demographic data on all of the cities in this study are from the United States Census Bureau. The Population Estimates Program publishes total resident population estimates and demographic components each year. The data in this report is for 2008. All these factors were found on the Fact Sheet section of each city in the United States Census Bureau website.

In order to analyze the data, one table was created combining all the information from both the FBI and the U.S. Census where each row represented one city and each column represents factual information about the city. This data table is included in the appendix.

Hypotheses

It is hypothesized that each of the factors in this study mentioned in the Research Problem section of this report has a statistically significant relationship with violent crime for the 40 cities in this study. With a confidence level set at 95%, it is hypothesized that many if not most of these factors have such a relationship.

Research Results

In this dataset, 40 cities were analyzed for violent crime. The average crime rate for these cities for 2008 was 943.1. Several criteria were used for selecting these cities. Cities had to be over 100,000 people and were preferred to be a significant part of the Metropolitan Statistical Area (M.S.A). The cities chosen had a population roughly between 100,000 and 1,000,000 with the exception of Albany and San Antonio.

Scatter plots were drawn for all thirteen factors in this study (See Appendix). Household Size, Percent High School Graduate or Higher, In labor force (population 16 years and older), Median age, and population were removed from analysis after visual inspection of the scatter plots because of little to no correlation seen. Multiple regression was carried out using Excel and SPSS for each the remaining nine factors. Table 1 shows the results of the Pearson Correlation

for each of these factors. Figure 1 and 2 shows that the violent crime data is very close to being normally distributed. As shown in Table 4, all of the factors fail to reject the null hypothesis where the confidence level was set at 95%. As it turns out, this study rather turned into a comparative study where each factor was compared with the next rather than indicating that any of them had a statistically significant relationship to violent crime. On the other hand, as shown in the ANOVA table (Table 3), the significance F value for this series is 0.00221. With a confidence level of 95%, this is well below the 0.05 that would be used to reject the null hypothesis which is that none of the factors affect violent crime. This is for the data as a whole though.

The top four factors in the order of their R^2 to violent crime are percent black (0.438), Median household Income (2008 Inflation adjusted Dollars) (0.357), Male, Now Married except separated (0.302), Individuals below poverty level (0.481), and Owner occupied housing units (0.162). The R^2 has a significant meaning. For example, according to this sample, 43.8 percent of the data shows a correlation with the percent black in a city and violent crime and 35.7 percent of the Median household income shows a correlation with violent crime.

As shown in table 1, Percent bachelors degree or higher R^2 (0.120) correlates with a reduction in violent crime significantly more than Percent high school graduate or higher (0.026). In fact, one the surprising findings in this report is how little high school graduation rate and violent crime rate correlate.

Table 1 - Selected Factors Correlated with Violent Crime

Factor	R	R^2
Violent Crime Rate per 100,000	1.000	1.000
Percent Black	0.662	0.438
Median household Income (2008 Inflation adjusted Dollars)	-0.597	0.357
Male, Now Married except separated	-0.550	0.302
Individuals below poverty level	0.481	0.232
Owner occupied housing units	-0.402	0.162
Percent Bachelors Degree or Higher	-0.346	0.120
Percent Hispanic or latino of any race	-0.344	0.118
Average Houshold Size	-0.334	0.111
Percent White	-0.318	0.101
Population	-0.271	0.074
In labor force (population 16 years and older)	-0.211	0.045
Percent High School Graduate or Higher	-0.161	0.026
Median Age	-0.063	0.004

Table 2 - Regression Statistics

Multiple R	0.789411
R Square	0.623169
Adjusted R Square	0.455689
Standard Error	271.4786
Observations	40

Table 3 - ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	12	3290751	274229.2	3.720853	0.002219969
Residual	27	1989917	73700.65		
Total	39	52806			

		68			

Table 4 – Regression analysis (Confidence Level of 95%)

	Coefficients	***Standard Error***	***t Stat***	***P-value***
Intercept	1500.045	3381.484	0.444	0.661
Percent High School Graduate or Higher	145.749	2638.933	0.055	0.956
Percent Bachelors Degree or Higher	-1333.251	1126.789	-1.183	0.247
Male, Now Married except separated	-698.469	1836.300	-0.380	0.707
In labor force (population 16 years and older)	1709.194	2143.706	0.797	0.432
Median household Income (2008 Inflation adjusted Dollars)	-0.019	0.013	-1.472	0.152
Individuals below poverty level	-2248.916	2708.287	-0.830	0.414
Owner occupied housing units	-982.070	1291.986	-0.760	0.454
Median Age	4.244	26.411	0.161	0.874
Percent White	326.924	618.178	0.529	0.601
Percent Black	1005.689	654.017	1.538	0.136
Percent Hispanic or Latino of any race	-129.270	603.532	-0.214	0.832
Population	0.000	0.000	0.919	0.366

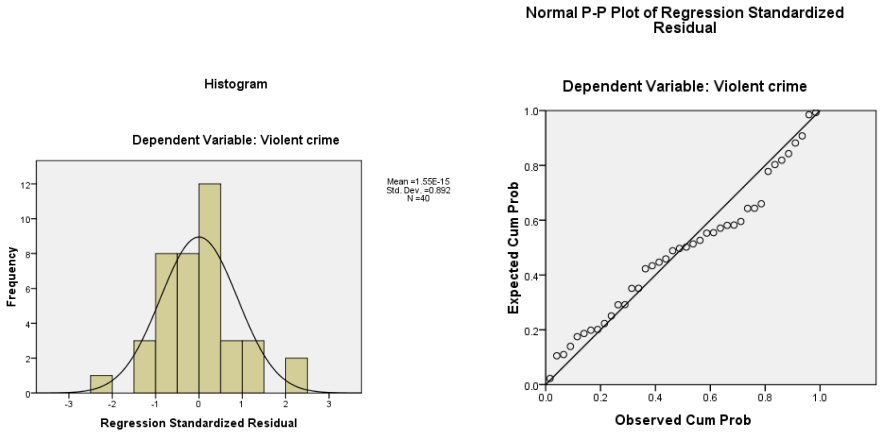


Figure1

Implications of the results for further research

The result of this study did not statistically relate these factors to violent crime in the strictest sense, but it does bring out differences among the factors. The finding that the percentage of the black population has the highest correlation to violent crime is very disturbing and more needs to be done in studying why this is the case. It is hypothesized that there is a high correlation with the percentage of blacks with other factors in this study such as median income, college graduation rate, the percentage of owner occupancy. Any policy maker serious about reducing the violent crime in his or her represented area needs look into how to get the black community in their area into college, home ownership, and resources for marriage and individual counseling. The median household income is the next highest factor that correlates to violent crime. Household income is strongly dependent on the ability of families to stay together and the education level of the household. More studies need to be done on the scope of this problem in cities with a high rate of violent crime. Property crime rate can also be analyzed by comparing these factors with the property crime rates for selected cities.

References

1. FBI, “2008 Crime in the United States by Metropolitan Statistical Area.” http://www.fbi.gov/ucr/cius2008/data/table_06.html.
2. U.S. Census Bureau, “American Fact Finder. Fact Sheet” http://factfinder.census.gov/home/saff/main.html?_lang=en.
3. Joseph Schmuller, PhD, *Statistical Analysis with Excel for Dummies* (Wiley Publishing, Inc., 2009), 282-289.
4. Frederick Williams and Peter Monge, Reasoning with Statistics (Thompson Wadsworth, 2001), 143-152.

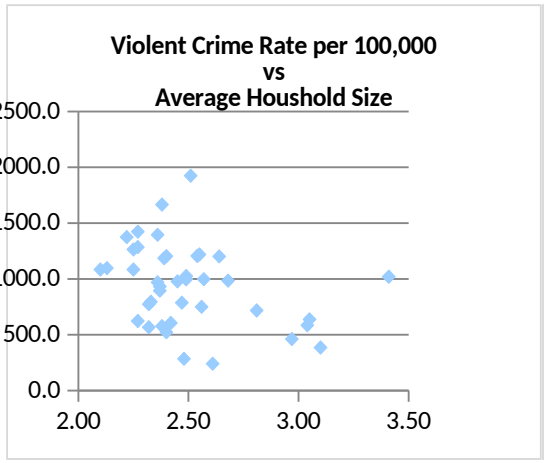
Appendix 1

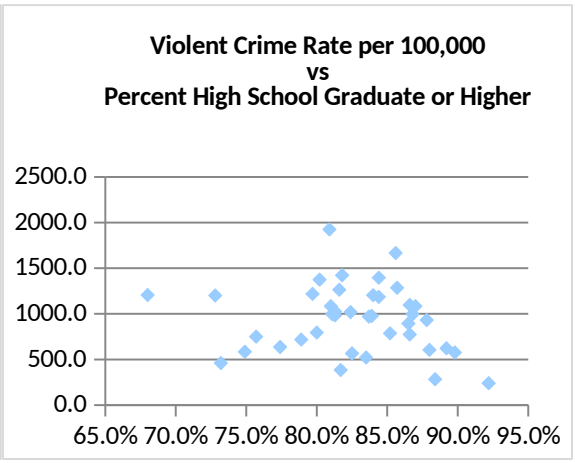
Data Table for Study

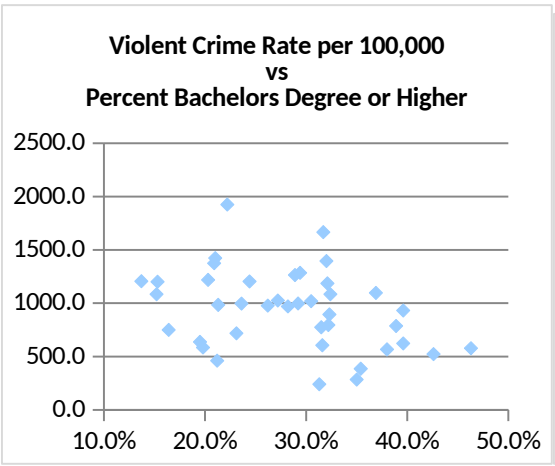
Counties/principal cities	Violent Crime Rate per 100,000	Average Household Size	Percent High School Graduate or Higher	Percent Bachelors Degree or Higher	Male, Now Married except separated	In labor force (population 16 years and older)	Median household Income (2008 Inflation adjusted Dollars)	Individuals below poverty level	Owner occupied housing units	Median Age	Percent White	Percent Black	Percent Hispanic or latino of any race	Population
City of Memphis	1925.02	2.51	80.9%	22.2%	35.5%	65.6%	$36,389	24.1%	54.5%	33.8	31.7%	62.6%	5.0%	672,046
City of Orlando	1666.17	2.38	85.6%	31.7%	39.7%	73.2%	$44,287	16.6%	42.1%	33.3	57.8%	27.4%	22.2%	229,808
City of Birmingham	1423.04	2.27	81.8%	21.0%	33.9%	62.0%	$32,070	24.2%	40.9%	37.1	23.0%	74.1%	2.9%	228,314
City of Nashville	1395.59	2.36	84.4%	32.0%	44.8%	68.2%	$46,029	16.3%	60.8%	36.2	64.9%	27.9%	7.8%	602,181
City of Buffalo	1374.82	2.22	80.2%	20.9%	32.0%	58.5%	$29,845	29.9%	42.9%	34.2	52.2%	39.1%	8.7%	270,289
City of Tulsa	1285.27	2.27	85.7%	29.4%	47.7%	66.3%	$39,373	19.2%	54.9%	35.8	68.5%	15.9%	12.1%	382,954
City of Cincinnati, OH	1263.95	2.25	81.6%	28.9%	31.0%	62.3%	$33,524	25.7%	42.6%	35.7	51.0%	44.4%	1.9%	332,608
City of Milwaukee	1218.84	2.55	79.7%	20.3%	35.5%	65.7%	$37,022	24.4%	48.4%	32.1	46.8%	38.3%	15.7%	602,131
City of Hartford	1206.16	2.54	68.0%	13.7%	28.3%	63.0%	$29,224	32.5%	25.7%	30.1	30.2%	37.2%	41.4%	124,610
City of Indianapolis	1204.34	2.40	84.0%	24.4%	45.9%	69.2%	$44,830	16.1%	60.3%	35.5	66.6%	25.9%	7.0%	808,329
City of Bridgeport	1201.52	2.64	72.8%	15.3%	40.1%	65.8%	$41,210	20.3%	46.6%	33.2	49.4%	34.1%	33.3%	136,327
City of Baton Rouge	1185.44	2.39	84.4%	32.1%	38.0%	64.8%	$37,242	25.7%	51.7%	30.3	40.8%	54.2%	2.3%	226,920
City of Albany	1097.16	2.13	86.6%	36.9%	28.6%	63.1%	$40,978	25.3%	40.9%	31.0	59.4%	28.7%	6.9%	94,152
City of Dayton	1084.18	2.25	81.0%	15.2%	33.2%	60.0%	$29,630	29.8%	50.9%	34.4	53.7%	42.7%	2.2%	154,218
City of Pittsburgh	1084.08	2.10	87.0%	32.4%	35.2%	59.5%	$34,834	21.6%	52.8%	36.7	67.0%	26.5%	1.8%	309,757
City of Grand Rapids	1025.91	2.49	81.3%	27.2%	42.0%	68.0%	$39,269	23.1%	40.9%	32.0	66.7%	20.3%	16.2%	193,096
City of New Orleans	1019.40	3.41	82.4%	30.5%	29.0%	59.5%	$39,398	23.8%	50.2%	31.0	34.4%	60.5%	4.5%	281,440
City of Sacramento	997.72	2.57	81.1%	29.2%	42.6%	65.0%	$50,651	15.3%	51.2%	33.2	51.1%	14.0%	24.6%	467,065
City of Jacksonville	996.43	2.49	86.8%	23.6%	49.9%	68.6%	$49,784	12.8%	64.1%	36.0	62.3%	30.1%	6.2%	806,080
City of Las Vegas Metropolitan Police Department	984.65	2.68	81.3%	21.3%	50.6%	66.8%	$55,113	11.7%	58.7%	35.5	70.8%	10.3%	30.0%	1,353,175
City of Oklahoma City	977.46	2.45	83.9%	26.2%	52.2%	67.4%	$43,011	16.5%	61.7%	35.0	67.7%	14.1%	14.6%	552,452
City of Worcester	969.79	2.36	83.7%	28.2%	38.0%	63.6%	$44,794	17.5%	46.7%	34.4	78.5%	9.0%	19.3%	177,151
City of Charlotte-Mecklenburg, NC	931.77	2.37	87.8%	39.6%	50.2%	74.0%	$53,125	12.3%	59.8%	34.6	55.5%	33.5%	11.2%	758,769
City of Albuquerque	894.47	2.37	86.5%	32.3%	47.3%	67.4%	$45,962	15.2%	61.7%	35.4	67.5%	3.5%	44.0%	527,464
City of Richmond	795.30	2.33	80.0%	32.2%	31.4%	63.1%	$38,385	23.0%	49.2%	35.5	42.4%	51.5%	4.6%	199,674
City of Salt Lake City	786.64	2.47	85.2%	38.9%	43.9%	70.8%	$44,552	16.8%	50.4%	31.6	80.7%	3.6%	21.9%	180,514
City of Columbus	774.19	2.32	86.6%	31.5%	41.1%	69.9%	$43,600	20.6%	50.6%	32.3	65.4%	26.4%	4.5%	751,887
City of Allentown, PA	749.99	2.56	75.7%	16.4%	41.4%	62.7%	$37,955	21.1%	50.9%	32.8	63.2%	11.0%	36.2%	107,335
City of San Antonio	717.78	2.81	78.9%	23.1%	48.4%	64.2%	$42,731	18.5%	59.7%	32.6	68.9%	6.6%	61.2%	1,351,244
City of Bakersfield	637.03	3.05	77.4%	19.5%	50.6%	66.0%	$52,160	16.5%	58.9%	29.1	57.4%	8.4%	42.2%	326,046
City of Portland, OR	622.94	2.27	89.2%	39.6%	43.8%	69.7%	$48,993	15.2%	57.7%	37.5	78.6%	6.4%	8.8%	553,023
City of Omaha, NE	605.62	2.42	88.0%	31.6%	46.6%	70.6%	$45,979	14.7%	60.0%	34.6	76.7%	12.8%	11.4%	437,238
City of Fresno	584.79	3.04	74.9%	19.8%	44.7%	63.9%	$42,616	23.7%	48.6%	28.8	53.3%	8.0%	44.6%	475,723
City of Raleigh	577.62	2.38	89.8%	46.3%	43.6%	71.5%	$53,926	12.8%	54.5%	32.2	60.2%	24.8%	9.7%	388,661
City of Denver	566.89	2.32	82.5%	38.0%	42.3%	69.5%	$45,002	18.7%	55.7%	35.3	74.1%	9.6%	34.1%	592,881
City of Austin	522.21	2.40	83.5%	42.6%	43.1%	74.1%	$51,004	17.6%	46.9%	31.4	63.7%	8.2%	35.5%	753,535
City of El Paso	461.32	2.97	73.2%	21.2%	53.2%	60.2%	$36,649	25.8%	62.5%	32.3	77.8%	3.1%	80.0%	612,374
City of San Jose	385.42	3.10	81.7%	35.4%	53.4%	67.8%	$79,796	9.7%	61.4%	35.6	48.6%	3.0%	31.5%	945,197
City of Honolulu	284.11	2.48	88.4%	35.0%	49.4%	62.3%	$59,194	10.2%	50.6%	42.9	21.4%	1.9%	3.8%	906,349
City of Virginia Beach, VA	240.23	2.61	92.2%	31.3%	54.6%	72.9%	$64,755	6.8%	68.0%	35.7	70.1%	19.4%	5.8%	434,163

Appendix 2

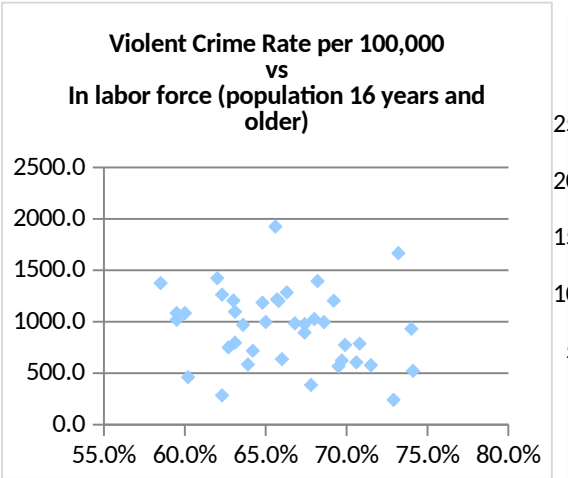
Scatter Plots

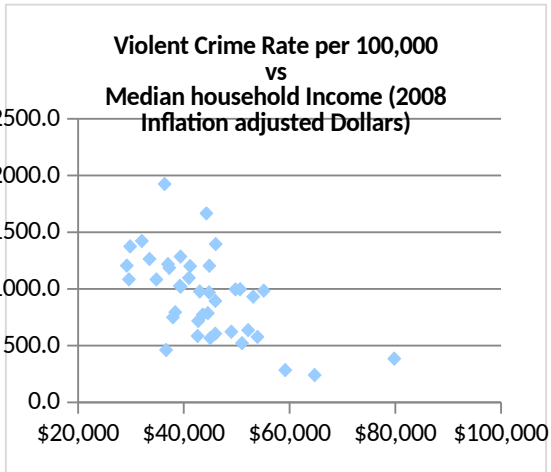


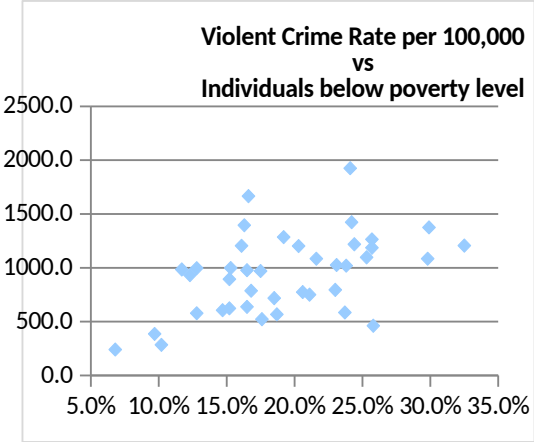


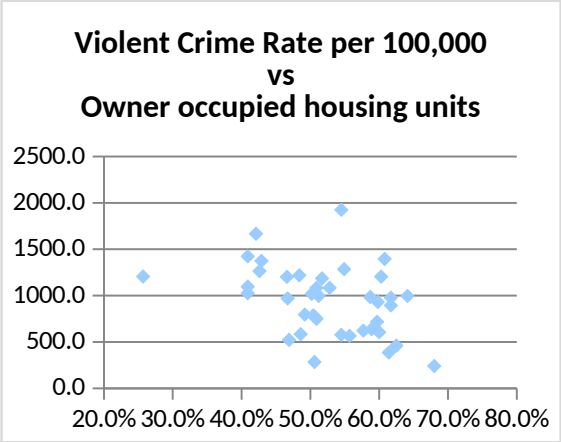


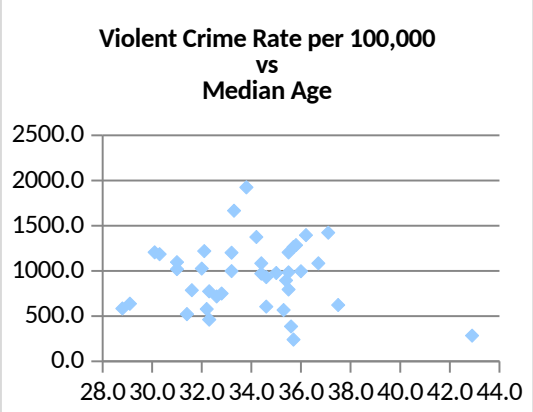


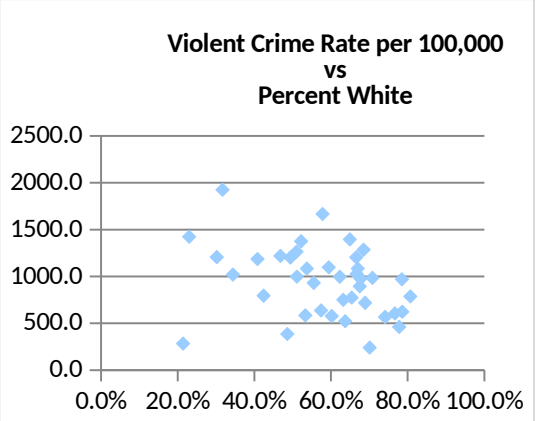


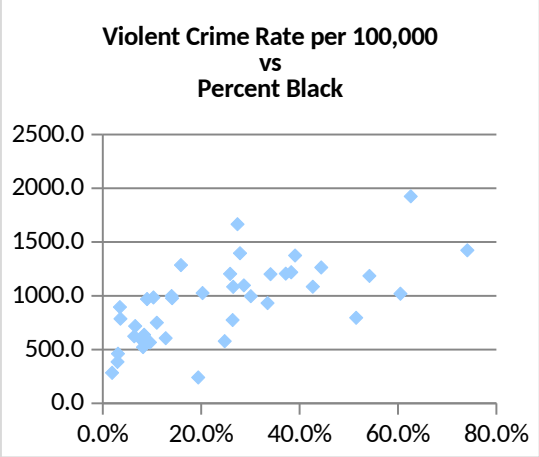


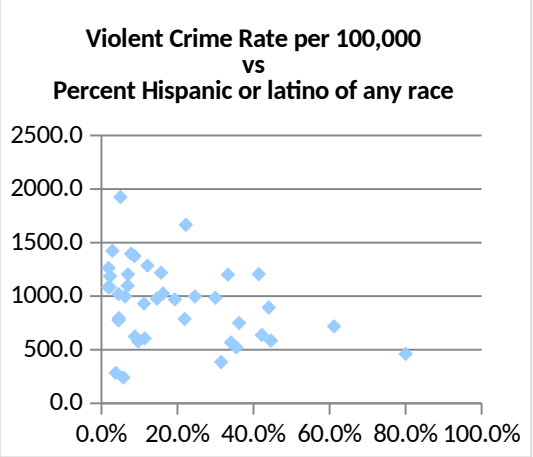

Violent Crime Rate per 100,000
vs
Individuals below poverty level
2500.0
2000.0
1500.0
1000.0
500.0
0.0
5.0% 10.0% 15.0% 20.0% 25.0% 30.0% 35.0%

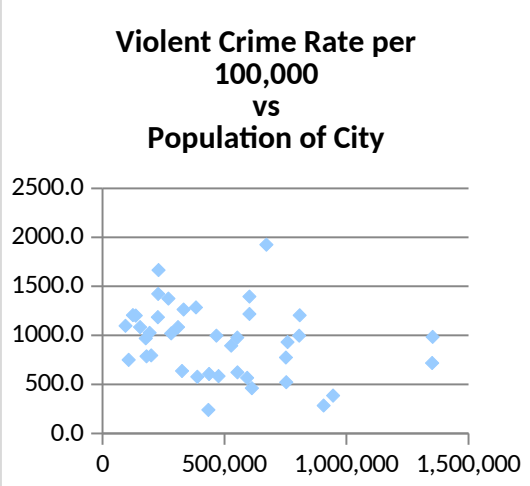

Violent Crime Rate per 100,000
vs
Owner occupied housing units
2500.0
2000.0
1500.0
1000.0
500.0
0.0
20.0% 30.0% 40.0% 50.0% 60.0% 70.0% 80.0%

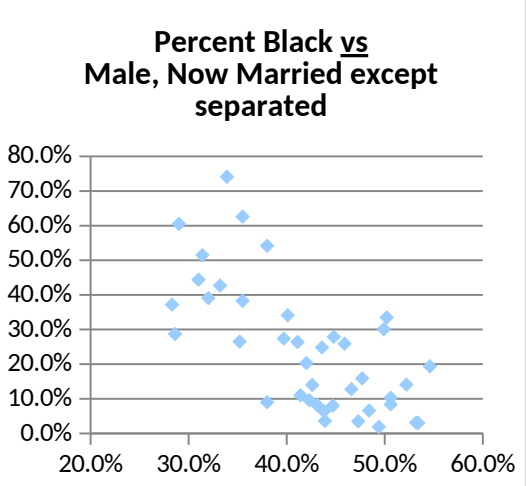

Violent Crime Rate per 100,000
vs
Median Age
2500.0
2000.0
1500.0
1000.0
500.0
0.0
28.0 30.0 32.0 34.0 36.0 38.0 40.0 42.0 44.0

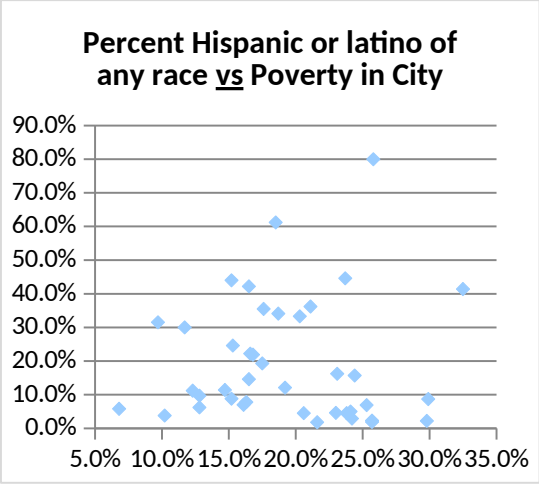

Violent Crime Rate per 100,000
vs
Percent White
2500.0
2000.0
1500.0
1000.0
500.0
0.0
0.0% 20.0% 40.0% 60.0% 80.0% 100.0%

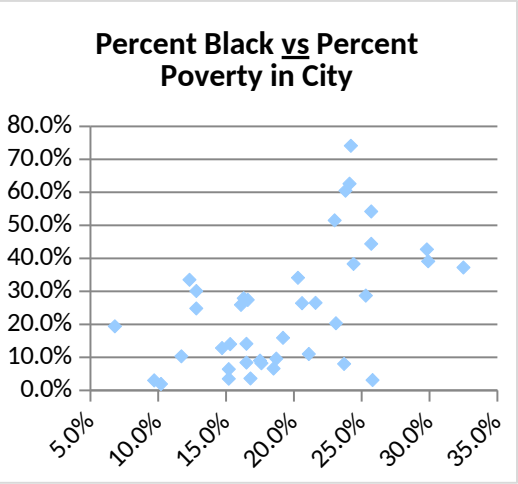

Violent Crime Rate per 100,000
vs
Percent Black
2500.0
2000.0
1500.0
1000.0
500.0
0.0
0.0% 20.0% 40.0% 60.0% 80.0%


Violent Crime Rate per 100,000
vs
Percent Hispanic or latino of any race
2500.0
2000.0
1500.0
1000.0
500.0
0.0
0.0% 20.0% 40.0% 60.0% 80.0% 100.0%


Violent Crime Rate per 100,000 vs Population of City
2500.0
2000.0
1500.0
1000.0
500.0
0.0
0
500,000
1,000,000
1,500,000


Percent Black vs Male, Now Married except separated
80.0%
70.0%
60.0%
50.0%
40.0%
30.0%
20.0%
10.0%
0.0%
20.0%
30.0%
40.0%
50.0%
60.0%


Percent Hispanic or latino of any race vs Poverty in City
90.0%
80.0%
70.0%
60.0%
50.0%
40.0%
30.0%
20.0%
10.0%
0.0%
5.0%
10.0%
15.0%
20.0%
25.0%
30.0%
35.0%


Percent Black vs Percent Poverty in City
80.0%
70.0%
60.0%
50.0%
40.0%
30.0%
20.0%
10.0%
0.0%
5.0%
10.0%
15.0%
20.0%
25.0%
30.0%
35.0%